

SAS Cortex Challenge

Donor Targeting & Donation Optimization

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Project Background

The Cortex Challenge aims to increase donations by using modeling and data analytics to identify potential donors and improve fundraising strategies.

Project Objective

This project uses predictive modeling and uplift analysis to identify which donors should be targeted in order to maximize total donations while considering campaign costs.

The Two-Stage Uplift Framework

⚙️ Why Two-Stage?

Business Logic:

"Should I donate?" (Probability)
"How much should I donate?" (Amount).

Data Logic:

Modeling the Probability and Amount components separately significantly enhances predictive accuracy.

☰ The 4 Score Data Tables

Two Situations:

If Contacted:

C-Give: Probability of donating after being contacted (P_c)

C-Amt: Expected donation amount after being contacted (M_c)

If Not Contacted:

NC-Give: Probability of making a donation when not contacted (P_{nc})

NC-Amt: Expected donation amount when not contacted (M_{nc})

☰ Model Performance Metrics

Calculation of Expected Donation Amount:

$$Expected\ Value = Probability \times Amount$$

Core Uplift Formula

$$Uplift = (P_c \times M_c) - (P_{nc} \times M_{nc})$$

© We do not contact everyone; we only contact the Persuadable.
Only when $Uplift > 5$ can we ensure that every outbound call generates positive Net Revenue.

Step 1 Probability to Give

 Pipeline – Gradient Model



 Model Statistics

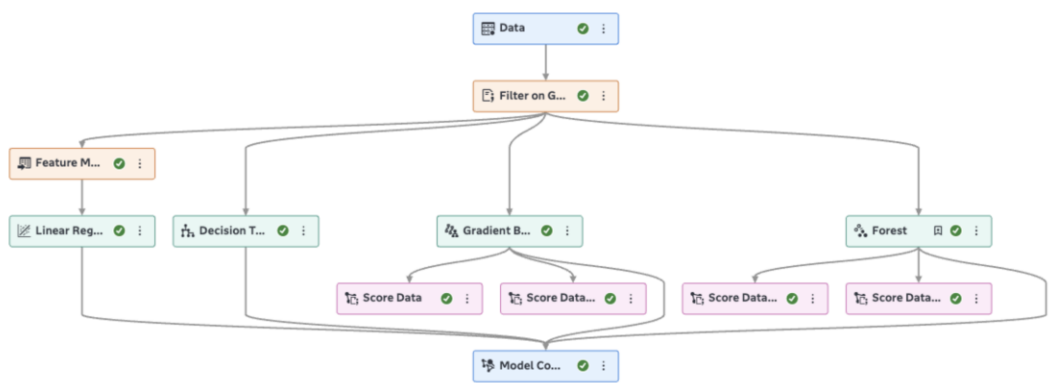
Fit Statistics

Statistics ...	Train: Forest	Validate: Forest	Test: Forest	Train: Gradient Boosting	Validate: Gradient Boosting
Area Under ROC	0.7814	0.6980	0.7096	0.7177	0.7052
Average Squared Error	0.1292	0.1455	0.1449	0.1422	0.1443
Divisor for ASE	60,000	29,999	10,001	60,000	29,999
Formatted Partition	1	0	2	1	0
Gamma	0.5769	0.4079	0.4314	0.4463	0.4208
Gini Coefficient	0.5628	0.3960	0.4193	0.4354	0.4104
KS (Youden)	0.4208	0.2818	0.3106	0.3144	0.2973
KS at User-Specified Cutoff	0.1630	0.0956	0.0920	0.1050	0.1064
KS Cutoff	0.2200	0.2100	0.1800	0.2300	0.1900
Misclassification Rate	0.1726	0.1923	0.1933	0.1893	0.1904
Misclassification Rate (Event)	0.1726	0.1923	0.1933	0.1893	0.1904

🎯 Goal: Predict the likelihood that a donor will make a donation

Step 2 Donation Amount

Pipeline – Gradient Model



Model Statistics

Fit Statistics

Statistics ...	Train: Forest	Validate: Forest	Test: Forest	Train: Gradient Boosting	Validate: Gradient Boosting	Test: Gradient Boosting
Average Squared Error	42,339.2150	49,563.3776	126,199.8566	48,841.0880	50,143.8133	127,187.3796
Divisor for ASE	12,259	6,071	1,985	12,259	6,071	1,985
Formatted Partition	1	0	2	1	0	2
Mean Absolute Error	61.2363	68.0625	80.5632	66.6486	68.1548	80.8862
Mean Squared Logarithmic Error	1.0678	1.3207	1.4103	1.2358	1.3227	1.4201
Number of Observations	12,259	6,071	1,985	12,259	6,071	1,985
Partition Indicator	1	0	2	1	0	2
R-Square	0.2550	0.0073	0.0133	0.1405	-0.0043	0.0056
Root Average Squared Error	205.7650	222.6283	355.2462	221.0002	223.9281	356.6334
Root Mean Absolute	7.8254	8.2500	8.9757	8.1639	8.2556	8.9937

🎯 Goal: Predict the expected donation amount given a donor will give

Players Ranking

Uploads Ranking

My Uploads

Interim

"Real Life"

Rank	Player	Operating surplus	Expenses	Donors contacted	Method	Uploads #	Selected
1	Prophet Margins	\$8,796,695.00	\$299,860.00	59972	level 2 xl	9	Selected
2	Coin Detective	\$8,791,995.00	\$300,000.00	60000	SUB 48	42	Selected
3	MI-Six Cortex	\$8,783,255.00	\$299,860.00	59972	dummmm	20	Selected
4	Banana Bread	\$8,772,025.00	\$300,000.00	60000	v8	10	Selected
5	EDDY O	\$8,769,440.00	\$299,860.00	59972	26th	29	Selected
6	Aditya Sakhale	\$8,743,645.00	\$300,000.00	60000	s121_g055	84	Selected

Operating Surplus



8,791,995

Expenses



300,000

Donors Contacted



60,000

Our models were able to **generalize well and avoid overfitting**, making the approach effective for real-world fundraising decisions